

ATD - Send Notifications - HK_HBAP - V1 -Carbon

Created by Unknown User (45224953), last modified by Kalman Su on Apr 24, 2024

Table of Contents

- Introduction
 - Scope
 - In Scope
 - Out of Scope
 - Channels
 - Assumptions & Constraints
 - References
 - Terms & Abbreviations
- Implementation Design
 - Downstream API
 - Business Logic
 - Technical Development
 - Data Mapping
 - Sample request header
 - Sample request payload
 - Sample request payload
 - Responses payload sample
 - Common Errors
 - Domain API - Error Codes Repository
 - Domain API - Implementation Guide for AIM Error Schema
 - listed some common validation errors
 - Derived Attributes
 - Request Header
 - Request Bodys
 - Responses
 - Errors
 - Domain API - Error Codes Repository
 - Domain API - Implementation Guide for AIM Error Schema
 - listed some common validation errors
 - Derived Attributes
- Deployment Design
 - Runtime Environments
- Key Technical Decisions

Revision History...

Version	Published	Changed By	Comment
CURRENT (v. 158)	Apr 24, 2024 14:54	 Kalman Su	
v. 157	Mar 14, 2024 17:29	 Kalman Su	
v. 156	Mar 13, 2024 10:26	 Kalman Su	
v. 155	Mar 12, 2024 17:18	 Kalman Su	
v. 154	Mar 11, 2024 10:20	 Kalman Su	
v. 153	Mar 07, 2024 10:54	 Kalman Su	
v. 152	Mar 06, 2024 16:16	 Kalman Su	
v. 151	Mar 06, 2024 15:50	 Kalman Su	
v. 150	Mar 06, 2024 15:44	 Kalman Su	
v. 149	Mar 05, 2024 15:22	 Kalman Su	
v. 148	Mar 01, 2024 09:58	 Kalman Su	
v. 147	Feb 29, 2024 09:29	 Kalman Su	
v. 146	Feb 26, 2024 16:00	 Woody Liu	
v. 145	Feb 08, 2024 11:43	 Kalman Su	
v. 144	Feb 08, 2024 11:13	 Kalman Su	
v. 143	Feb 06, 2024 11:56	 Woody Liu	
v. 142	Feb 02, 2024 15:13	 Richard Song_SP	
v. 141	Jan 31, 2024 10:57	 Woody Liu	
v. 140	Jan 30, 2024 18:09	 Kalman Su  Woody Liu	
v. 139	Jan 26, 2024 17:31	 Woody Liu	
v. 138	Jan 26, 2024 17:24	 Woody Liu	
v. 137	Jan 26, 2024 15:24	 Kalman Su	
v. 136	Jan 26, 2024 09:39	 Kalman Su	
v. 135	Jan 25, 2024 18:38	 Kalman Su	
v. 134	Jan 24, 2024 20:01	 Woody Liu  Maggie X Y Li_SP	
v. 133	Dec 15, 2023 13:15	 Richard Song_SP	

Document Information

Status	Owner	Authors	Version	Reviewed By	Approved By
RELEASE	@ Unknown User (45224953)	@ Unknown User (45224953)	1.050	TPE Forum 22 Dec 2021	TPE Forum 22 Dec 2021
RELEASE	@ Richard Song_SP	@ Richard Song_SP	1.051	Roger 11 Oct 2023	Roger 11 Oct 2023
APPROVED	@ Maggie X Y Li_SP	@ Maggie X Y Li_SP	1.052	Roger 08 Nov 2023	Roger 08 Nov 2023
DRAFT	@ Richard Song_SP	@ Richard Song_SP	1.053	Roger	Roger
DRAFT	@ Richard Song_SP	@ Richard Song_SP	1.054		
RC	@ Woody Liu	@ Woody Liu	1.055		
RELEASE	@ Kalman Su	@ Kalman Su	1.056	Lending Designer Forum	Lending Designer Forum

About this Document:

A Technical Design describes the implementation and deployment variation for a specific market or feature. For the API Contract Specification and API Product Design details, refer to links in the API Information section below.

API Information

Name	notifications--zz
Layer	PROCESS
Version	1.050
Domain	Keep me secure
Subdomain	Protect me from fraudulent / unfair activity
Function	Authorize a transaction on my account
Product	
Increment	 DSDOM-156879 - AMH MWS - Deliver eAdvice in SMS and email DONE  DSDOM-290000 - RM FX 2023 - Perform FX transaction DONE  DSDOM-347248 - HK - HKMA Credit Limit Utilisation Notification DONE  DSDOM-382814 - NDA-ed International VS Proj - PAPI to ICCM FUNNEL  DSDOM-381225 - HKMA A3b - Positive consent for Credit Limit Increase (CLI) - PAPI ENGINEERING  DSDOM-396431 - HK-API changes to accommodate ICCM to MDP migration for appointment booking in APAC region ON HOLD
Roadmap	
Repository	to be filled by develop team
Contract	https://alm-github.systems.uk.hsbc/domain-papi-ramls/notifications--zz/tree/v1-carbon
Model	
Replaces	
Source	https://alm-github.systems.uk.hsbc/domain-papi/notifications--zz

Version	Published	Changed By	Comment
v. 132	Dec 15, 2023 13:14	 Richard Song_SP	
v. 131	Dec 06, 2023 18:37	 Richard Song_SP	
v. 130	Dec 06, 2023 18:34	 Richard Song_SP	
v. 129	Nov 29, 2023 14:58	 Maggie X Y Li_SP	
v. 128	Nov 29, 2023 14:48	 Maggie X Y Li_SP	
v. 127	Nov 29, 2023 14:15	 Maggie X Y Li_SP	
v. 126	Nov 24, 2023 16:12	 Maggie X Y Li_SP	
v. 125	Nov 22, 2023 09:44	 Maggie X Y Li_SP	
v. 124	Nov 21, 2023 14:22	 Maggie X Y Li_SP	
v. 123	Nov 09, 2023 10:57	 Richard Song_SP	
v. 122	Nov 09, 2023 10:53	 Richard Song_SP	
v. 121	Nov 08, 2023 18:03	 Richard Song_SP	
v. 120	Nov 08, 2023 09:52	 Maggie X Y Li_SP	
v. 119	Nov 08, 2023 09:42	 Maggie X Y Li_SP	
v. 118	Nov 08, 2023 09:37	 Maggie X Y Li_SP	
v. 117	Nov 08, 2023 09:33	 Maggie X Y Li_SP	
v. 116	Nov 08, 2023 09:28	 Maggie X Y Li_SP	
v. 115	Nov 07, 2023 11:42	 Maggie X Y Li_SP	
v. 114	Nov 07, 2023 11:41	 Maggie X Y Li_SP	
v. 113	Nov 07, 2023 11:38	 Maggie X Y Li_SP	
v. 112	Nov 07, 2023 11:36	 Maggie X Y Li_SP	
v. 111	Nov 06, 2023 15:51	 Maggie X Y Li_SP	
v. 110	Nov 02, 2023 19:00	 Richard Song_SP	
v. 109	Nov 02, 2023 18:53	 Richard Song_SP	
v. 108	Nov 02, 2023 18:50	 Richard Song_SP	
v. 107	Nov 02, 2023 18:42	 Richard Song_SP	
v. 106	Nov 02, 2023 15:30	 Richard Song_SP	
v. 105	Nov 02, 2023 15:04	 Richard Song_SP	
v. 104	Nov 02, 2023 14:53	 Richard Song_SP	
v. 103	Nov 02, 2023 10:19	 Richard Song_SP	
v. 102	Nov 02, 2023 10:18	 Richard Song_SP	
		 Maggie X Y Li_SP	
v. 101	Nov 01, 2023 14:59	 Richard Song_SP	
v. 100	Nov 01, 2023 14:35	 Richard Song_SP	
v. 99	Nov 01, 2023 14:07	 Richard Song_SP	
v. 98	Nov 01, 2023 13:53	 Richard Song_SP	
v. 97	Nov 01, 2023 12:43	 Richard Song_SP	
v. 96	Nov 01, 2023 12:42	 Richard Song_SP	
v. 95	Nov 01, 2023 12:40	 Richard Song_SP	
v. 94	Nov 01, 2023 11:43	 Maggie X Y Li_SP	
v. 93	Oct 31, 2023 20:12	 Richard Song_SP	
v. 92	Oct 31, 2023 17:22	 Maggie X Y Li_SP	
v. 91	Oct 20, 2023 16:12	 Maggie X Y Li_SP	
v. 90	Oct 20, 2023 16:12	 Maggie X Y Li_SP	

Version	Published	Changed By	Comment
v. 89	Oct 20, 2023 16:10	 Maggie X Y Li_SP	
v. 88	Oct 20, 2023 16:07	 Maggie X Y Li_SP	
v. 87	Oct 20, 2023 14:56	 Maggie X Y Li_SP	
v. 86	Oct 19, 2023 10:56	 Maggie X Y Li_SP	
v. 85	Oct 19, 2023 10:49	 Maggie X Y Li_SP	
v. 84	Oct 18, 2023 11:03	 Maggie X Y Li_SP	
v. 83	Oct 18, 2023 10:45	 Maggie X Y Li_SP	
v. 82	Oct 18, 2023 10:42	 Maggie X Y Li_SP	
v. 81	Oct 17, 2023 11:32	 Maggie X Y Li_SP	
v. 80	Oct 17, 2023 10:46	 Maggie X Y Li_SP	
v. 79	Oct 12, 2023 16:00	 Maggie X Y Li_SP	
v. 78	Oct 06, 2023 14:30	 Maggie X Y Li_SP	
v. 77	Oct 05, 2023 18:01	 Maggie X Y Li_SP	
v. 76	Oct 05, 2023 17:59	 Maggie X Y Li_SP	
v. 75	Oct 05, 2023 17:55	 Maggie X Y Li_SP	
v. 74	Oct 05, 2023 17:54	 Maggie X Y Li_SP	
v. 73	Sep 14, 2023 11:13	 Richard Song_SP	
v. 72	Sep 14, 2023 11:13	 Richard Song_SP	
v. 71	Aug 29, 2023 17:15	 Richard Song_SP	
v. 70	Aug 28, 2023 18:50	 Richard Song_SP	
v. 69	Aug 28, 2023 17:14	 Richard Song_SP	
v. 68	Aug 28, 2023 16:53	 Richard Song_SP	
v. 67	Aug 28, 2023 16:43	 Richard Song_SP	
v. 66	Aug 25, 2023 19:27	 Richard Song_SP	
v. 65	Aug 25, 2023 15:55	 Richard Song_SP	
v. 64	Aug 25, 2023 14:48	 Roger Y W Pan	
v. 63	Aug 25, 2023 14:47	 Roger Y W Pan	
v. 62	Aug 25, 2023 14:29	 Richard Song_SP	
v. 61	Aug 24, 2023 16:37	 Richard Song_SP	
v. 60	Aug 23, 2023 19:22	 Richard Song_SP	
v. 59	Aug 23, 2023 19:06	 Richard Song_SP	
v. 58	Aug 23, 2023 18:17	 Richard Song_SP	
v. 57	Aug 23, 2023 10:14	 Richard Song_SP	
v. 56	Aug 17, 2023 19:02	 Richard Song_SP	
v. 55	Aug 16, 2023 11:48	 Richard Song_SP	
v. 54	Aug 15, 2023 14:46	 Richard Song_SP	
v. 53	Aug 15, 2023 14:27	 Richard Song_SP	
v. 52	Aug 14, 2023 17:44	 Richard Song_SP	
v. 51	Aug 14, 2023 17:27	 Richard Song_SP	
v. 50	Aug 08, 2023 18:02	 Richard Song_SP	
v. 49	Aug 08, 2023 17:46	 Richard Song_SP	
v. 48	Aug 08, 2023 14:14	 Richard Song_SP	
v. 47	Aug 08, 2023 14:13	 Richard Song_SP	
v. 46	Aug 08, 2023 12:40	 Richard Song_SP	

Version	Published	Changed By	Comment
v. 45	Aug 07, 2023 20:03	 Richard Song_SP	
v. 44	Aug 07, 2023 19:51	 Richard Song_SP	
v. 43	Aug 07, 2023 15:36	 Richard Song_SP	
v. 42	Aug 07, 2023 15:34	 Richard Song_SP	
v. 41	Aug 07, 2023 15:30	 Richard Song_SP	
v. 40	Aug 07, 2023 10:25	 Richard Song_SP	
v. 39	Aug 04, 2023 12:28	 Richard Song_SP	
v. 38	Aug 03, 2023 18:30	 Richard Song_SP	
v. 37	Aug 03, 2023 18:07	 Richard Song_SP	
v. 36	Aug 03, 2023 18:02	 Richard Song_SP	
v. 35	Aug 02, 2023 19:27	 Richard Song_SP	
v. 34	Aug 02, 2023 19:19	 Richard Song_SP	
v. 33	Aug 02, 2023 14:53	 Richard Song_SP	
v. 32	Aug 02, 2023 14:33	 Richard Song_SP	
v. 31	Aug 02, 2023 13:04	 Richard Song_SP	
v. 30	Aug 01, 2023 15:01	 Richard Song_SP	
v. 29	Aug 01, 2023 15:00	 Richard Song_SP	
v. 28	Aug 01, 2023 14:55	 Richard Song_SP	
v. 27	Aug 01, 2023 14:54	 Richard Song_SP	
v. 26	Aug 01, 2023 14:52	 Richard Song_SP	
v. 25	Aug 01, 2023 14:51	 Richard Song_SP	
v. 24	Aug 01, 2023 14:49	 Richard Song_SP	
v. 23	Aug 01, 2023 11:48	 Richard Song_SP	
v. 22	Jul 31, 2023 16:15	 Richard Song_SP	
v. 21	Jul 31, 2023 15:30	 Richard Song_SP	
v. 20	Jul 31, 2023 13:01	 Richard Song_SP	
v. 19	Jul 31, 2023 12:46	 Richard Song_SP	
v. 18	Jul 31, 2023 12:08	 Richard Song_SP	
v. 17	Jul 26, 2023 19:07	 Richard Song_SP	
v. 16	Jul 26, 2023 18:08	 Richard Song_SP	
v. 15	Jul 25, 2023 17:27	 Richard Song_SP	
v. 14	Jul 25, 2023 14:58	 Richard Song_SP	
v. 13	Jul 25, 2023 14:47	 Richard Song_SP	
v. 12	Jul 25, 2023 14:46	 Richard Song_SP	
v. 11	Jul 25, 2023 14:44	 Richard Song_SP	
v. 10	Jul 25, 2023 14:40	 Richard Song_SP	
v. 9	Jul 25, 2023 14:34	 Richard Song_SP	
v. 8	Jul 25, 2023 14:25	 Richard Song_SP	
v. 7	Mar 29, 2023 11:57	 Wilson W X Gao	
v. 6	Feb 10, 2023 15:22	 Woody Liu	Reverted from v. 5
v. 5	Oct 24, 2022 12:14	 Woody Liu	
v. 4	May 11, 2022 17:21	 Unknown User (45224953)	
v. 3	May 06, 2022 16:53	 Unknown User (45224953)	

Version	Published	Changed By	Comment
v. 2	May 06, 2022 16:53	 Unknown User (45224953)	
v. 1	May 05, 2022 17:38	 Unknown User (45224953)	

Introduction

MyWorkSpace (MWS) is the strategy platform for staff application, with the e-advice is provided, branch staff will be able to deliver the transaction advice in SMS and email. for the sending notification function, MWS will call this PAPI to involve ICCM to send out the notification.

 [DSDOM-347248](#) - HK - HKMA Credit Limit Utilisation Notification DONE

HCC will trigger this API to send email, SMS or both. But for this scenario, ICCM will not be triggered directly, MDP will be triggered instead, which is the strategic solution for message delivery.

 [DSDOM-382550](#) - HK - HKMA Credit Limit Utilisation Push Notification DONE

HCC will trigger this API to send PUSH notification.(v1.053)

Scope

In Scope

This technical design supports the following:

- send email notification to customer
- send SMS notification to customer
- send both EMAIL and SMS notification to customer in one go via MDP  [DSDOM-347248](#) - HK - HKMA Credit Limit Utilisation Notification DONE (v1.051)
- send push notification to customer mobile app via MDP  [DSDOM-382550](#) - HK - HKMA Credit Limit Utilisation Push Notification DONE (v1.053)

Out of Scope

The following is out of scope of this technical design:

Channels

Channel	Scope
WEB	
STAFFWEB	
TABLET	
STAFFTAB	
MOBILE	
KIOSK	
POS	
MOBMSG	
CHAT	
ATM	
OPENBANK	
VIDEO	
VOICE	
PARTNER	

Assumptions & Constraints

The following assumptions have been made about this API:

- Each time, only one delivery method (email or SMS) need to proceed
- when notificationModeType is MDP_NOTIFICATION, multiple delivery channels are supported in one go.

References

Review the following documents for additional information:

Ref ID	Document Reference
solution design	Hong Kong e-Advice

Terms & Abbreviations

The following terms and abbreviations are used in this document:

Term / Abbreviation	Description
SAPI	System API
PAPI	Process API
MyWorkSpace	MWS

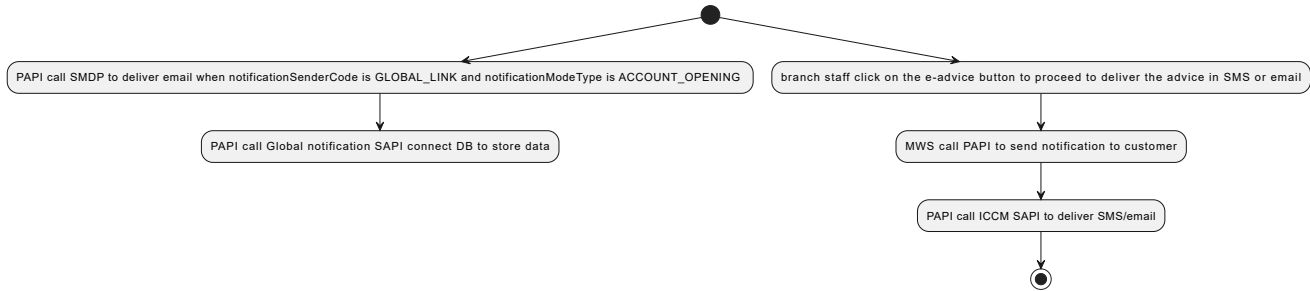
Implementation Design

Downstream API

Service Name	Service Type	JIRA Ticket	Description	API Contract
ICCM online email services	SYSTEM API			wpb-iccm-online-email-services - ICCM Product Details - Confluence (uk.hsbc)
ICCM online SMS services	SYSTEM API			wpb-iccm-composeonlinesms-sa - ICCM Product Details - Confluence (uk.hsbc)
Message delivery platform	SYSTEM API			2. Getting Started#SMS
customers-info--obs-hk	SYSTEM API			https://alm-github.systems.uk.hsbc/system-api/customers-info--obs-hk/blob/V4-PROD/src/main/api/customers-info--obs-hk.raml
SMDP email services	PROCESS API		to send mail to customer when notificationModeType is ACCOUNT_OPENING and notificationSenderCode is GLOBAL_LINK	2. Getting Started#2.GettingStarted-2.GettingStarted-2.6.2DataDrivenAPISpec
Global notification SAPI	SYSTEM API		after send mail to customer through SMDP services then call global notification SAPI to connect DB to store data for tracking send status.	DB Details SAPI Design
MDP 2.0	SYSTEM API			TBC

Business Logic

The following logic is performed by the API:

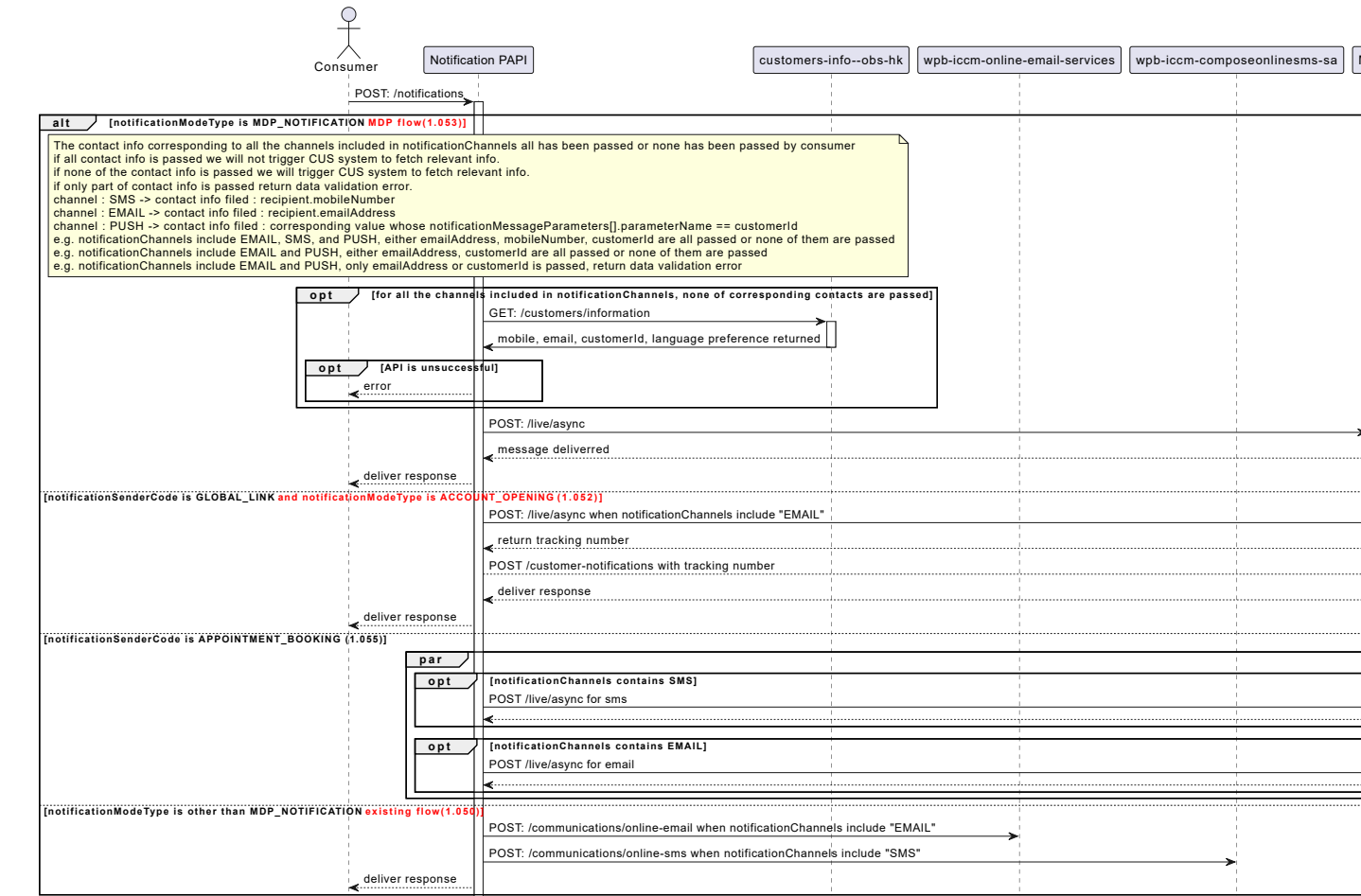


DSDOM-347248 - HK - HKMA Credit Limit Utilisation Notification DONE

1.051/1.052

DSDOM-382550 - HK - HKMA Credit Limit Utilisation Push Notification DONE

Current



Technical Development

The following functional changes are required in this API increment:

Function Name	Change Type	Change Description	JIRA Ticket
/notifications	MODIFY	- add enum value "EADVICE" to field "notificationModeType" - add enum value "MWS" to field "notificationSenderCode"	DSDOM-168847 - MWS - Deliver eAdvice in SMS and email DONE
/notifications	MODIFY	- add enum value "MDP_NOTIFICATION" to field "notificationModeType" - add enum value "HCC" to field "notificationSenderCode"	
/notifications	MODIFY	- add enum value "ACCOUNT_OPENING" to field "notificationModeType" - add enum value "GLOBAL_LINK" to field "notificationSenderCode"	DSDOM-382814 - NDA-ed International VS Proj - PAPI to ICCM FUNNEL
/notifications	MODIFY	- add enum value "PUSH" to field "notificationChannels" - support PUSH notification	DSDOM-383094 - notifications--zz POST /notifications HK-HBAP HKMA credit limit utilization push notification DONE
/notifications v1.054	MODIFY	Encrypt the AccountIdentifier field of the CLIPreApprovalNotification use case request body fields, because the calling system is sending plain card number whereas consuming system requires encrypted card number.	

Function Name	Change Type	Change Description	JIRA Ticket
/notifications	MODIFY	scale to support simply booking project	 DSDOM-396431 - HK-API changes to accommodate ICCM to MDP migration for appointment booking in APAC region ON HOLD
/notifications v1.056	MODIFY	scale to support 3DS notification (only support SMS and receive mobile number from CUS)	 DSDOM-401527 - notifications--zz POST /notifications HK-HBAP 3DS BUILD COMPLETE

Data Mapping

☐ POST /notifications when notificationSenderCode is APPOINTMENT_BOOKING_NOTIFICATIONS

MDP

Requests

 SAPI request payload

as MDP team requested, they only use the flat JSON mode to pass request payload, in this mode, the request payload is a JSON contains fixed message meta data and the parameter to fill message template, e.g.

```
{  "phoneNumber": "1234",  ...  "template_field_1": "abcdef",  ...}
```

Since (e.g. API Version)	Type	Attribute	Downstream Attribute Type	Downstream Attribute	Description / Mapping logic
1.055		n/a	HEADER	Content-Type	set to "application/json"
1.055	HEADER	X-HSBC-Source-System-Id	HEADER	X-HSBC-Source-System-Id	The EIM Application Instance Id of the source system (or application) at HSBC that initiates the call to an API. Value will be the EIM code of the PAPI
1.055	HEADER	X-HSBC-Chnl-CountryCode	HEADER	X-HSBC-Chnl-CountryCode	The ISO 3166-1 alpha-2 country code of the channel application (or system) that was used by a customer (or member of staff) to interact with HSBC.
1.055	HEADER	X-HSBC-Chnl-Group-Member	HEADER	X-HSBC-Chnl-Group-Member	The HSBC Group member that the channel belongs to
1.055	HEADER	X-HSBC-GBGF	HEADER	X-HSBC-GBGF	The Global Business or Global Function (GBGF) as defined by the Business
1.055	HEADER	X-HSBC-Global-Channel-Id	HEADER	X-HSBC-Global-Channel-Id	This header contains the name of the channel that initiated the call to the first API. Sample = PARTNER
1.055	HEADER	X-Forwarded-For	HEADER	X-HSBC-Src-Device-Id	mapped to X-Forwarded-For, IP address of customer device
1.055	HEADER	X-HSBC-Src-UserAgent	HEADER	X-HSBC-Src-UserAgent	User Agent from user's machine/browser
1.055	HEADER	X-HSBC-Request-Correlation-Id	HEADER	X-HSBC-Request-Correlation-Id	a UUID for each request that is generated by the originating client application
1.055	HEADER	X-HSBC-E2E-Trust-Token	HEADER	X-HSBC-E2E-Trust-Token	Contains Identity information about the authenticated caller. JWT will be created using external B2B pattern.
1.055			HEADER	X-HSBC-Messaging-UseCase	required for eb2b, use case will be use to determine the template portfolio.
1.055	HEADER	X-HSBC-Chnl-CountryCode	HEADER	X-HSBC-Messaging-TenantId	required for MDP 2.0. set to "AU-HBAU-WPB", as discussed with MDP, this will be a suitable granularity.
1.055			HEADER	X-HSBC-Messaging-AppAlias	optional
1.055			HEADER	X-HSBC-Messaging-Category	optional
1.055		informationClassificationType		n/a	for email, possible values are "PUBLIC" "INTERNAL" "RESTRICTED" "HIGHLY_RESTRICTED"
1.055			PAYLOAD	PAYLOAD_UUID	generated by PAPI for each request send to SAPI. Unique Identifier for a single record in payload.
1.055	PAYLOAD	notificationChannels	HEADER	X-HSBC-Messaging-Channels	PAPI value to SAPI value mapping SMS - sms EMAIL - email
1.055	PAYLOAD	notificationModeType		N/A	following 4 types are used by this journey "CONFIRM", "CANCEL", "MODIFY", "REMIND"
1.055	PAYLOAD	notificationSenderCode		N/A	consumer to pass "APPOINTMENT_BOOKING_NOTIFICATIONS"
1.055	PAYLOAD	recipient.mobileNumber	PAYLOAD	phoneNumber	required when notificationChannels include "SMS"
1.055	PAYLOAD	recipient.emailAddress	PAYLOAD	emailAddress	required when notificationChannels include "EMAIL"
1.055	PAYLOAD	languagePreferenceCode	PAYLOAD	smsLanguage	required when notificationChannels include "SMS" PAPI to maintain the mapping, currently we only map "en_GB" to "en"
1.055	PAYLOAD	languagePreferenceCode	PAYLOAD	emailLanguage	required when notificationChannels include "EMAIL" PAPI to maintain the mapping, currently we only map "en_GB" to "en"
1.055	PAYLOAD	notificationMessageParameters ["emailDetails.subject"]	PAYLOAD	emailSubject	optional. client application my derive the mail subject when template can't provide a desired subject
1.055			PAYLOAD	Sms_TemplateId Email_TemplateId	to determine what template will be used. depend on use case

Since (e.g. API Version)	Type	Attribute	Downstream Attribute Type	Downstream Attribute	Description / Mapping logic
1.055	PAYLOAD	notificationMessageParameters[]	PAYLOAD	<body>[key]:[value]	rest fields in notificationMessageParameters, refer above MDP payload structure information
1.055	PAYLOAD	notificationMessageParameters[], parameterName	PAYLOAD	<body>[key]:[value]	refer above MDP payload structure information
1.055	PAYLOAD	notificationMessageParameters[], parameterValue	PAYLOAD	<body>[key]:[value]	refer above MDP payload structure information

Responses

PAPI Attribute	SAPI Attribute	Comment / Description / Mapping logic
notificationsDetails[].notificationChannel	DERIVED	"SMS" or "EMAIL" base on the SAPI called
notificationsDetails[].notificationIdentifier	DERIVED	corresponding value of SAPI field PAYLOAD_UUID
notificationsDetails[].requestReceivedDateTime	N/A	generate current timestamp in UTC format

Errors - errors in grey are handled by common lib

Since (e.g. API Version)	HTTP Status Code	Error Type	Downstream Error Type / Code / BRC	Description
1.1	400	TERR_30001	C0000 C0006 - C0008 C0010 - C0013	Bad request at downstream API
1.1	400	TERR_30001	http status: 413	Bad request at downstream API C0003: Over the limits of number of records per API Call C0004: Over the limits of size of records per API Call
1.1	500	SERR_10002	401	Invalid Client Id or Secret Key to the Downstream API
1.1	500	SERR_10003	403	Authentication Failure in Downstream API
1.1	500	TERR_30004	403 I0001	should not happen if configuration are correct incorrect X-HSBC-Messaging-UseCase
1.1	500	TERR_30004	500 S0000 - S0003	Technical Error in Downstream API
1.1	500	TERR_30004	415	Unsupported Media Type
1.1	500	TERR_30003	429	Too Many Request

Sample request header

request header

X-HSBC-Client-Id:XXX
X-HSBC-Chnl-CountryCode:HK
X-HSBC-Chnl-Group-Member:HBAP
X-HSBC-Global-Channel-Id:PARTNER
X-HSBC-GBGF:WPB
X-HSBC-Request-Correlation-Id:2d933d80-be43-11e7-92e1-005056a61740
X-HSBC-Src-UserAgent:Mozilla/5.0 (iPad; U; CPU OS 3_2_1 like Mac OS X; en-us) AppleWebKit/531.21.10 (KHTML, like Gecko) Mobile/7B405
X-HSBC-Source-System-Id:9000091
Accept-Language:en_GB

Sample request payload

SMS request sample

```
{
  "notificationModeType": "REMIND",
  "informationClassificationType":"PUBLIC",
  "recipient":{
    "mobileNumber":"+86 10010010100"
  },
  "languagePreferenceCode":"en_GB",
  "notificationSenderCode": "APPOINTMENT_BOOKING_NOTIFICATIONS",
  "notificationChannels": [ "SMS" ],
  "notificationMessageParameters": [
    {
      "parameterName": "Appointment_reference",
      "parameterValue": "2980"
    },
    {
      "parameterName": "Sales_staff_name",
      "parameterValue": "SBKHKJRNUSER1"
    },
    {
      "parameterName": "linkForRescheduleCancel",
      "parameterValue": "https://studio.bookingbug.com/?client=hsbc-gba&environment=stagingpurchase=0aDgt8sM_P3ynY-hMjknNg&config.bbLocaleSettings.defaultLanguage=en#/confirm"
    },
    {
      "parameterName": "APPT_DT_GENDATE",
      "parameterValue": "2021-06-25 - HKT"
    },
    {
      "parameterName": "APPT_TM_GENTIME",
      "parameterValue": "13:00:00 PM -- HKT"
    }
  ]
}
```

Sample request payload

Email request sample

```
{
  "notificationModeType": "CONFIRM",
  "informationClassificationType":"PUBLIC",
  "recipient":{
    "emailAddress":"SBKHKJRNUSER1@hsbc.com"
  },
  "languagePreferenceCode":"en_GB",
  "notificationSenderCode": "APPOINTMENT_BOOKING_NOTIFICATIONS",
```



```
"notificationChannels": [ "EMAIL" ],
"notificationMessageParameters": [
  {
    "parameterName": "emailDetails.fromEmailAddress",
    "parameterValue": "iccmtesting@mail.test.hsbc.com.hk"
  },
  {
    "parameterName": "emailDetails.subject",
    "parameterValue": "Meeting Invitation with SBKHKJRNUSER1 on 25 Jun 2021 at 13:00"
  },
  {
    "parameterName": "emailDetails.sender",
    "parameterValue": "SBKHKJRNUSER1"
  },
  {
    "parameterName": "Appointment_reference",
    "parameterValue": "2980"
  },
  {
    "parameterName": "Sales_staff_name",
    "parameterValue": "SBKHKJRNUSER1"
  },
  {
    "parameterName": "linkForRescheduleCancel",
    "parameterValue": "https://studio.bookingbug.com/?client=hsbc-gba&environment=stagingpurchase=0aDgt8sM_P3ynY-hMjKxNg&config.bbLocaleSettings.defaultLanguage=en#/confirm"
  },
  {
    "parameterName": "APPT_DT_GENDATE",
    "parameterValue": "2021-06-25 - HKT"
  },
  {
    "parameterName": "APPT_TM_GENTIME",
    "parameterValue": "13:00:00 PM - HKT"
  }
]
}
```

Responses payload sample

```
Response payload sample

{
  "notificationsDetails": [
    {
      "notificationChannel": "SMS",
      "notificationIdentifier": "2113f68d-88d6-450d-8169-f39cfc5645cb",
      "requestReceivedDateTime": "2017-10-23T18:25:43.511Z"
    }
  ]
}
```

Common Errors

[Domain API - Error Codes Repository](#)

[Domain API - Implementation Guide for AIM Error Schema](#)

listed some common validation errors

PAPI HTTP Status Code	PAPI Error Code	PAPI Error Description
403	BERR_20001	Request data business validation error outside Hystrix
400	BERR_20002	Request data validation error, <u>use case: when channel is provided but related contact is not presented</u>
400	BERR_20003	Request data constraint validation error
400	BERR_20004	Request data type validation error
400	BERR_20005	Request data type validation error after controller
400	BERR_20006	Request data cannot be bind to model
400	BERR_20007	Request data is in invalid format
403	BERR_20008	Request data business validation error in Hystrix

Derived Attributes

X-HSBC-Source-System-Name

X-HSBC-GBGF + X-HSBC-Channel-Id + X-HSBC-Source-System-Id

X-HSBC-Language-Preference-Code

PAPI maintains an mapping to translate from Accept-Language in request header. eg. "en_GB" => "ENG"

X-HSBC-iccategory

PAPI received	"PUBLIC"	"INTERNAL"	"RESTRICTED"	"HIGHLY_RESTRICTED"
SAPI REQUIRED	1	2	3	4

CommunicationDetails.templateDescription

in current system LST2.GEN.GBAHKEMAIL_REMINDENG

This field is the key to find our what template should be used in downstream.

if (notificationSenderCode == "APPOINTMENT_BOOKING_NOTIFICATIONS" && X-HSBC-Chnl-CountryCode == "HK" && X-HSBC-Chnl-Group-Member == "HBAP")

then:{

if (notificationMessageParameters[].parameterName=="templateDescription" is not null)

then: (templateDescription(ICCM) = notificationMessageParameters[].parameterValue(notificationMessageParameters[] parameterName=="templateDescription"))

else:{

templateDescription(ICCM) = "LST2.GEN." + notificationModeType + "_" + notificationSenderCode(value to concat template description)

notificationSenderCode(body)	notificationSenderCode(value to concat template description)
APPOINTMENT_BOOKING_NOTIFICATIONS	APPT_BKG

}

}

○ POST /notifications when notificationModeType is MDP_NOTIFICATION

GET /customers/information		MDP							
Requests									
Since (e.g. API Version)	Type	Attribute	Downstream Attribute Type	Downstream Attribute	Description / Mapping logic				
	HEADER	X-HSBC-Chnl-CountryCode	HEADER	X-HSBC-Chnl-CountryCode					
	HEADER	X-HSBC-Chnl-Group-Member	HEADER	X-HSBC-Chnl-Group-Member					
	HEADER	X-HSBC-GBGF	HEADER	X-HSBC-GBGF					
		DERIVED	HEADER	Content-Type	application/json				
		DERIVED	HEADER	X-HSBC-Messaging-UseCase	corresponding value whose notificationMessageParameters[].parameterName == messagingUseCase				
			HEADER	X-HSBC-Messaging-Channels					
	HEADER	X-HSBC-E2E-Trust-Token	HEADER	X-HSBC-E2E-Trust-Token	iB2B				
	HEADER	X-HSBC-Request-Correlation-Id	HEADER	X-HSBC-Request-Correlation-Id					
	HEADER	X-HSBC-Session-Correlation-Id	HEADER	X-HSBC-Session-Correlation-Id					
	PAYLOAD	notificationModeType	PAYLOAD	NA	MDP_NOTIFICATION				
	PAYLOAD	notificationSenderCode	PAYLOAD	NA	HCC, MWS, THIRD_PARTY(added in v1.056)				
	PAYLOAD	notificationChannels[]	HEADER	X-HSBC-Messaging-Channels	possible values of PAPI ["SMS", "EMAIL", "PUSH"] PUSH is added in v1.053 only pass this header when messagingUseCase = InAppAuth3DSForHK(added in v1.056) <table><tr><td>PAPI</td><td>SAPI</td></tr><tr><td>SMS</td><td>sms</td></tr></table>	PAPI	SAPI	SMS	sms
PAPI	SAPI								
SMS	sms								
		DERIVED	PAYLOAD	phoneNumber	Need to add to the SAPI key/value object ({"phoneNumber": value(please refer to next row)})				
	PAYLOAD	recipient.mobileNumber	PAYLOAD		if not provided, will fetch from CUS (customerInformationDetails.formattedMobilePhoneNumber) If the calling system intends to retrieve such info from CUS, then they should not pass such info, since if such info is passed, it will be ignored. If mobile number is invalid, please pass "", will only happen when SMS and EMAIL channel are both needed				
		DERIVED	PAYLOAD	emailAddress	Need to add to the SAPI key/value object ({"emailAddress": value(please refer to next row)})				
	PAYLOAD	recipient.emailAddress	PAYLOAD		if not provided, will fetch from CUS If the calling system intends to retrieve such info from CUS, then they should not pass such info, since if such info is passed, it will be ignored. If email address is invalid, please pass "", will only happen when SMS and EMAIL channel are both needed				
	PAYLOAD	languagePreferenceCode	PAYLOAD	NA	If EMAIL/SMS/EMAIL and SMS are provided, this value should also be passed and format should be in en-HK, if not pass If EMAIL/SMS/EMAIL and SMS are not provided, this value will not be used, since this API will fetch such info from CUS.				
					format of SAPI request, so for each notificationMessageParameters element, value of parameterName serves as key and pairs to the json object. ({"Key":"Value","Key1":"Value1","Key2":"Value2"})				
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	PAYLOAD_UUID	If the calling system wants to query the delivery status of SMS/EMAIL, they need to send this field. if it is blank, PAPI will generate UUID and pass to MDP and return as response				
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD						
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	messagingUseCase	messagingUseCase, if no such key or corresponding value is blank, data validation error Please remove this key/pair after setting the value to header				
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		messagingUseCase defined in MDP				
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	AccountIdentifier	when messagingUseCase is equal to CLIPreApprovalNotification, if no such key or corresponding value is blank, data validation error v1.054				
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		consumer will send plain card number, PAPI needs to encrypt before passing it to SAPI. e.g. customer is sending notificationMessageParameters[0].parameterName = AccountIdentifier, notificationMessageParameters[0].parameterValue = VEdTRINCu0dTQINHEpD8bvH6QSGR7fv1.054				
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	customerId	if EMAIL/ SMS/ EMAIL and SMS are provided, in this case, we will not trigger CUS, thus this customerId should also be passed to CUS. else manually add the customerId and corresponding value to the SAPI json object:				
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		if the contact info corresponding to all the channels included in notificationChannels has not been passed, take customerInformationDetails.internalCustomerId else this should be passed by consumer				
v1.056		N/A	PAYLOAD	consumerName	if messagingUseCase == InAppAuth3DSForHK { if mobileNumber startsWith +852, use HBAP_HK_3DS_CC_CSL				

Since (e.g. API Version)	Type	Attribute	Downstream Attribute Type	Downstream Attribute	Description / Mapping logic
					otherwise use HBAP_HK_3DS_CC_SAP }
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	smsLanguage	Need to add to the SAPI key/value object This key-value pair is only applicable when notificationChannels contains SMS
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		if EMAIL/ SMS / EMAIL and SMS are provided, in this case, we will not trigger CUS, get the language from languagePrefi set to en-HK. Otherwise get it from customerInformationDetails.preferWrittenLanguage if not null and size > 0, pick the first one if preferWrittenLanguage is C, then zh-HK if preferWrittenLanguage is E, then en-HK(if use case is CreditLimitIncreaseOffer, please pass en) ① Due to communication gap, introducing this special handling for CreditLimitIncreaseOffer use case if empty, default to en-HK
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	emailLanguage	Need to add to the SAPI key/value object This key-value pair is only applicable when notificationChannels contains EMAIL
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		please refer to logic of smsLanguage
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	base64EncodedFields	Specific field for HCC, due to historical encoding issues, HCC cannot send clear traditional Chinese, so need to base64 e and corresponding value is not blank, PAPI needs to split the corresponding value by "," and get all the keys which require And each key, PAPI needs to check if notificationMessageParameters contain such key and replace corresponding value After checking and replacing, please remove this key and value from notificationMessageParameters.
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD		pseudo code <pre>private void base64DecodeAndReplace(List<NotificationMessageParameter> notificationMessageP if (notificationMessageParameters == null notificationMessageParameters.size() < return; } NotificationMessageParameter base64EncodedField = notificationMessageParameters .stream().filter(notificationMessageParameter -> notificationMessageParameter.parameter .findFirst() .orElse(new NotificationMessageParameter()); if(StringUtils.isBlank(base64EncodedField.parameterValue)) { return; } Base64.Decoder decoder = Base64.getDecoder(); notificationMessageParameters.forEach(notificationMessageParameter -> { if (base64EncodedField.parameterValue.contains(notificationMessageParameter.para notificationMessageParameter.parameterValue = new String(decoder.decode(notificationMessageParameter.parameterValu }) }); notificationMessageParameters.remove(base64EncodedField); }</pre>
	PAYLOAD	notificationMessageParameters[].parameterName	PAYLOAD	keyN	Other dynamic key/value pairs used in AEM template. if both email and sms are supported, key/value pairs should contain the keys used in email and sms are different, if the keys are the same, the value should be the same as well.
	PAYLOAD	notificationMessageParameters[].parameterValue	PAYLOAD	valueN	NOTE: Upon receiving the request, PAPI needs to iterate notificationMessageParameters array and check if parameterVa AMHHCC_DEFAULT_EMPTY_STRING_PLACEHOLDER, if yes, need to replace this AMHHCC_DEFAULT_EMPTY_! Reason of doing so is HCC need to pass empty parameterValue but existing logic of this API does not allow empty.

Responses

PAPI Attribute	SAPI Attribute	Comment / Description / Mapping logic
notificationsDetails[].notificationIdentifier	PAYLOAD_UUID	return corresponding value whose notificationMessageParameters[].parameterName == PAYLOAD_UUID
notificationsDetails[].notificationChannel		SMS or EMAIL or PUSH, please conform to request notification channel
notificationsDetails[].requestReceivedDateTime		generate current timestamp in UTC format

Errors

HTTP Status Code	Error Type	Downstream Error Type / Code / BRC	Description

○ POST/notifications when notificationSenderCode is GLOBAL_LINK and notificationModeType is ACCOUNT_OPENING

SMDP email service

Global notification SAPI

SMDP endpoint

POST

live/async

Request Header

Type	PAPI Attribute	SAPI Attribute	Description / Mapping logic
HEADER	Content-Type	Content-Type	application/json
HEADER	X-HSBC-E2E-Trust-Token	X-HSBC-E2E-Trust-Token	Contains Identity information about the authenticated caller. JWT will be created using iB2B pattern.
HEADER	X-HSBC-Chnl-CountryCode	X-HSBC-Chnl-CountryCode	Static value: "HK"

Type	PAPI Attribute	SAPI Attribute	Description / Mapping logic
HEADER	X-HSBC-Chnl-Group-Member	X-HSBC-Chnl-Group-Member	Static value: "HBAP"
DERIVED	notificationMessageParameters[].parameterValue	X-HSBC-Messaging-UseCase	WHEN notificationMessageParameters[i] . parameterName = "notificationJourney"
DERIVED	notificationChannels	X-HSBC-Messaging-Channels	lower case "notificationChannels" - email
DERIVED	N/A	X-HSBC-Messaging-TenantId	should hardcode:INTL

Request Bodys

Type	PAPI Attribute	Type	SAPI Attribute	Description / Mapping logic
BODY	notificationModeType	BODY	N/A	ACCOUNT_OPENING
BODY	identifierType	BODY	N/A	CUSTOMER_IDENTIFIER
BODY	notificationSenderCode	BODY	N/A	GLOBAL_LINK
BODY	notificationChannels	BODY	N/A	EMAIL
HEADER	X-HSBC-Request-Correlation-Id	BODY	PAYLOAD_UUID	
BODY	identifierValue	BODY	CustomerId	
BODY	recipient.emailAddress	BODY	emailAddress	
BODY	notificationMessageParameters[].parameterValue	BODY	emailLanguage	WHEN notificationMessageParameters[i] . parameterName = "languagePreferredCode"
BODY	notificationMessageParameters[].parameterValue	BODY	referralId	WHEN notificationMessageParameters[i] . parameterName = "referralId"
BODY	notificationMessageParameters[].parameterValue	BODY	firstName	WHEN notificationMessageParameters[i] . parameterName = "firstName"
BODY	notificationMessageParameters[].parameterValue	BODY	lastName	WHEN notificationMessageParameters[i] . parameterName = "lastName"
BODY	notificationMessageParameters[].parameterValue	BODY	fromEmailAddress	WHEN notificationMessageParameters[i] . parameterName = "fromEmailAddress"
BODY	notificationMessageParameters[].parameterValue	BODY	emailSender	WHEN notificationMessageParameters[i].parameterName = "emailSender"
BODY	notificationMessageParameters[].parameterValue	BODY	emailSubject	WHEN notificationMessageParameters[i].parameterName = "emailSubject"
BODY	notificationMessageParameters[].parameterValue	BODY	templateDescription	WHEN notificationMessageParameters[i] . parameterName = "templateId"

Responses

PAPI Attribute	SAPI Attribute	Comment / Description / Mapping logic
notificationsDetails[].notificationChannel	DERIVED	PAPI set "EMAIL"
notificationsDetails[].notificationIdentifier	DERIVED	X-HSBC-Request-Correlation-Id
notificationsDetails[].requestReceivedDateTime	N/A	generate current timestamp in UTC format

Error

HTTP Response Code	PAPI Error Code	SAPI Error Code / BRC	Description
401	BERR_20003	C0001	Access Denied (AWS_IAM / E2ETrust Validation Failed)
400	BERR_20003	C0006	X-HSBC-Chnl-CountryCode or X-HSBC-Chnl-Group-Member is empty
400	BERR_20003	C0007	Invalid Request Body
400	BERR_20003	C0008	Invalid Request Parameters
400	BERR_20003	C0010	UseCase Information is mandatory for Real Time Messaging
400	BERR_20003	C0011	PAYLOAD_UUID is mandatory for every record
400	BERR_20003	C0012	Invalid UseCase Format
400	BERR_20003	C0013	Invalid Entity Found
400	BERR_20003	C0000 I0001	Other Bad Request Error
413	TERR_30011	C0003	Over the limits of number of records per API Call
413	TERR_30011	C0004	Over the limits of size of records per API Call
415	TERR_30011	C0009	Unsupported Media Type
429	TERR_30011	C0005	Too many concurrent API Calls (Throttled)
500	TERR_30004	S0001	Configuration Error
500	TERR_30004	S0002	Receiving Batch Records failed
500	TERR_30004	S0003	Failed Connecting to Backend Service
500	TERR_30004	S0000	Other Backend Error

Request Header

Type	PAPI Attribute	SAPI Attribute	Description / Mapping logicX-HSBC-Locale
DERIVE		Content-Type	set to "application/json"
	X-HSBC-Source-System-Id	X-HSBC-Source-System-Id	The EIM Application Instance Id of the source system (or application) at HSBC that initiates the call to an API. Value will be the EIM code of the PAPI
	X-HSBC-Chnl-CountryCode	X-HSBC-Chnl-CountryCode	The ISO 3166-1 alpha-2 country code of the channel application (or system) that was used by a customer (or member of staff) to interact with HSBC.
	X-HSBC-Chnl-Group-Member	X-HSBC-Chnl-Group-Member	The HSBC Group member that the channel belongs to
	X-HSBC-GBGF	X-HSBC-GBGF	The Global Business or Global Function (GBGF) as defined by the Business
	X-HSBC-Global-Channel-Id	X-HSBC-Channel-Id	This header contains the name of the channel that initiated the call to the first API. Sample = PARTNER
	X-Forwarded-For	X-HSBC-Src-Device-Id	mapped to X-Forwarded-For, IP address of customer device
	X-HSBC-Src-UserAgent	X-HSBC-Src-UserAgent	User Agent from user's machine/browser
	X-HSBC-Request-Correlation-Id	X-HSBC-Request-Correlation-Id	a UUID for each request that is generated by the originating client application
	X-HSBC-client-id	X-HSBC-client-id	The client id obtained from the API manager. This client id must have been registered to use this API by requesting access to it. This needs to be generated by engineering team. this is just for testing stage.
	X-HSBC-client-secret	X-HSBC-client-secret	The client id obtained from the API manager. This client id must have been registered to use this API by requesting access to it. This needs to be generated by engineering team. this is just for testing stage.
DERIVE		X-HSBC-E2E-Trust-Token	Contains Identity information about the authenticated caller. JWT will be created using external B2B pattern.
BODY	languagePreferenceCode	X-HSBC-Language-Preference-Code	PAPI to maintain the mapping, currently we use mapping as below
			languagePreferenceCodeX-HSBC-Language-Preference-CodenotificationChannels
			bil_hkBILemail
			en_hkENGSMSSMS
			zh_hkCHISMS
DERIVE		X-HSBC-Source-System-Name	WSPC
		X-HSBC-Deliverable-Id	AMH_WORKSPACE
		X-HSBC-Deliverable-Type	NTF
BODY	informationClassificationType	X-HSBC-iccategory	Optional in ICCM, default type is RESTRICTED mapping from PAPI to ICCM
			"PUBLIC"→"1", "INTERNAL"→"2", "RESTRICTED"→ "3", "HIGHLY_RESTRICTED"→ "4"

Request Bodys

ICCM SAPI/Contract details:

online email: [wpb-iccm-online-email-services](#) - [ICCM Product Details](#) - [Confluence \(uk.hsbc\)](#)

online sms: [wpb-iccm-composeonlinesms-sa](#) - [ICCM Product Details](#) - [Confluence \(uk.hsbc\)](#)

Type	PAPI Attribute	SAPI Attribute	Description / Mapping logic
	notificationModeType		Enum Type; value to concat template EADVICE (used in v1.050)
	identifierType		N/A
	identifierValue		N/A
	notificationSenderCode		Enum Type; value to concat template MWS (used in v1.050)
	notificationChannels		Enum Type to determine which SAPIs to call when SMS and EMAIL have very different template for on notification type
DERIVE	recipient.mobileNumber	smsDetails.mobileNumber	required when notificationChannels include "SMS"
DERIVE	recipient.emailAddress	emailDetails.recipientEmailAddress	required when notificationChannels include "EMAIL"
N/A	notificationMessageParameters ["emailDetails.fromEmailAddress"]	emailDetails. fromEmailAddress	optional example: the mail address of who triggered the appointment change hardcode
N/A	notificationMessageParameters ["emailDetails.sender"]	emailDetails. sender	optional example: the staff name of who triggered the appointment change hardcode
DERIVE	notificationMessageParameters ["emailDetails.subject"]	emailDetails. subject	optional. client application my derive the mail subject when template can't provide a desired subject hardcode
DERIVE	notificationMessageParameters[].parameterValue	mergeFieldList[].name	other fields in notificationMessageParameters
DERIVE	notificationMessageParameters[].parameterName	mergeFieldList[].value	other fields in notificationMessageParameters
BODY		communicationDetail/s. templateDescription	We propose the template to be apply under pattern like "LST2.GEN." + notificationModeType + "_" + notificationMessageParameters["transactionType"] + "_" +notificationSenderCode example: LST2.GEN.EADVICE_XXX_MWS
DERIVE		communicationDetail/s. segmentIdentifierCustomerGroup	set to 1000, currently don't have special requirement here
N/A		communicationDetails. customerIdentificationNumber	it's for ICCM side to have historic search, but it's optional and they haven't implement this function yet.
N/A		communicationDetails. communicationCategory	It's for ICCM downstream system, and can be generated by ICCM
N/A		communicationDetails. contentUrl	It's for special business maintained template, not used in this case
N/A		encrytedFieldList[]	not used in current requirement

Sample request header:

```
requestHeaders
X-HSBC-Client-Id:XXX
X-HSBC-Chnl-Country-Code:GB
X-HSBC-Chnl-Group-Member:HBEU
X-HSBC-Global-Channel-Id:PARTNER
X-HSBC-GBGF:WPB
```

X-HSBC-Request-Correlation-Id:2d933d80-be43-11e7-92e1-005056a61740
X-HSBC-Src-UserAgent:Mozilla/5.0 (iPad; U; CPU OS 3_2_1 like Mac OS X; en-us) AppleWebKit/531.21.10 (KHTML, like Gecko) Mobile/7B405
X-HSBC-Source-System-Id:9000091
Accept-Language:en_hk

Sms Request Sample

```
{
  "notificationModeType": "EADVICE",
  "recipient":{
    "mobileNumber":"+86 10010010100"
  },
  "languagePreferenceCode":"en_hk",
  "notificationSenderCode": "MWS",
  "notificationChannels": [ "SMS" ],
  "notificationMessageParameters": [
    {
      "parameterName": "transactionType",
      "parameterValue": "XXX"
    },
    {
      "parameterName": "linkForRescheduleCancel",
      "parameterValue": "https://studio.bookingbug.com/?client=hsbc-gba&environment=stagingpurchase=0aDgt8sM_P3ynY-hMjkkNg&config.bbLocaleSettings.defaultLanguage=en#/confirm"
    },
    {
      "parameterName": "APPT_DT_GENDATE",
      "parameterValue": "2021-06-25 - HKT"
    },
    {
      "parameterName": "APPT_TM_GENTIME",
      "parameterValue": "13:00:00 PM - HKT"
    }
  ]
}
```

Sample request payload:

Email request sample

```
{
  "notificationModeType": "EADVICE",
  "infomationClassificationType":"PUBLIC",
  "recipient":{
    "emailAddress":"SBKHKJRNUSER1@hsbc.com"
  },
  "languagePreferenceCode":"en_GB",
  "notificationSenderCode": "MWS",
  "notificationChannels": [ "EMAIL" ],
  "notificationMessageParameters": [
    {
      "parameterName": "transactionType",
      "parameterValue": "XXX"
    },
    {
      "parameterName": "emailDetails.fromEmailAddress",
      "parameterValue": "icmtesting@mail.test.hsbc.com.hk"
    },
    {
      "parameterName": "emailDetails.subject",
      "parameterValue": "Meeting Invitation with SBKHKJRNUSER1 on 25 Jun 2021 at 13:00"
    },
    {
      "parameterName": "emailDetails.sender",
      "parameterValue": "SBKHKJRNUSER1"
    },
    {
      "parameterName": "Appointment_reference",
      "parameterValue": "2980"
    },
    {
      "parameterName": "Sales_staff_name",
      "parameterValue": "SBKHKJRNUSER1"
    },
    {
      "parameterName": "linkForRescheduleCancel",
      "parameterValue": "https://studio.bookingbug.com/?client=hsbc-gba&environment=stagingpurchase=0aDgt8sM_P3ynY-hMjkkNg&config.bbLocaleSettings.defaultLanguage=en#/confirm"
    },
    {
      "parameterName": "APPT_DT_GENDATE",
      "parameterValue": "2021-06-25 - HKT"
    },
    {
      "parameterName": "APPT_TM_GENTIME",
      "parameterValue": "13:00:00 PM - HKT"
    }
  ]
}
```

Responses

PAPI Attribute	SAPI Attribute	Comment / Description / Mapping logic
notificationsDetails[].notificationChannel		"SMS"/"EMAIL" base on the SAPI called
notificationsDetails[].notificationIdentifier	ResponseDetails.trackingNumber	denotes tracing number for the submitted request
notificationsDetails[].requestReceivedDateTime	ResponseDetails.requestRecievedDateTime	Denotes request received date and time

Response payload Sample:

Response Payload Sample

```
{
  "notificationsDetails": [
    {
      "notificationChannel": "SMS",
      "notificationIdentifier": "2113f68d-88d6-450d-8169-f39cfc5645cb",
      "requestReceivedDateTime": "2017-10-23T18:25:43.511Z"
    }
  ]
}
```

```
}
    ]
}
```

Errors

HTTP Response Code	PAPI Error Code	SAPI Error Code / BRC	Description
400	TERR_30001	OES4001	PAPI: Bad request at downstream API SAPI: Request body validation failure
400	TERR_30001	OES4002	PAPI: Bad request at downstream API SAPI: Request header validation failure
500	TERR_30004	OES5000 - OES5006	PAPI: Technical Error in Downstream API SAPI: Unable to call back end service or other exception when SAPI calling it's downstream APIs

Domain API - Error Codes Repository

Domain API - Implementation Guide for AIM Error Schema

listed some common validation errors

PAPI HTTP Status Code	PAPI Error Code	PAPI Error Description
403	BERR_20001	Request data business validation error outside Hystrix
400	BERR_20002	Request data validation error, <u>use case: when channel is provided but related contact is not presented</u>
400	BERR_20003	Request data constraint validation error
400	BERR_20004	Request data type validation error
400	BERR_20005	Request data type validation error after controller
400	BERR_20006	Request data cannot be bind to model
400	BERR_20007	Request data is in invalid format
403	BERR_20008	Request data business validation error in Hystrix

Derived Attributes

X-HSBC-Source-System-Name

X-HSBC-GBGF + X-HSBC-Channel-Id + X-HSBC-Source-System-Id

X-HSBC-Language-Preference-Code

PAPI maintains an mapping to translate from Accept-Language in request header, eg. "en_GB" => "ENG"

X-HSBC-iccategory

PAPI received	"PUBLIC"	"INTERNAL"	"RESTRICTED"	"HIGHLY_RESTRICTED"
SAPI REQUIRED	1	2	3	4

CommunicationDetails. templateDescription

in current system LST2.GEN.GBAHKEMAIL_REMINDENG

propose the template to be apply under pattern like

notificationModeType + notificationSenderCode + <Header.Language-Preference-Code> + notificationChannels

example:

LST2.GEN.GBAHK_APPOINTMENT_CONFIRM_JRNI_ENG_EMAIL

LST2.GEN.GBAHK_APPOINTMENT_CONFIRM_JRNI_ENG_SMS

engineer team will register the templates in this pattern and provide notificationModeType to client application

Deployment Design

Please note: as confirmed with business, SMS deployment will be not in scope on Jun.16 2022 release

Runtime Environments

Dev team is expected to fill this table after development is complete

This API is deployed to the following environments:

Environment Name	Gateway URL	Backing Service Base URIs
CERT		
PROD		

Key Technical Decisions

The following Key Technical Decisions have been made during the technical design of this API:

Id	Raised On	Raised By	Description & Options	Priority	Status	Decision & Rationale	Agreed By	Agreed On
1								
2								
3								